

Hien Ohnaka

Doctoral Student & Engineer

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Profile

Doctoral student at The University of Osaka and full-time engineer at LY Corporation, Japan.

Experience

LY Corporation , Full-time Engineer	Japan Apr 2026 – present 7 months
LY Corporation R&D , Part-time Researcher - Improved phonemic and prosodic annotation models for text-to-speech; the work was accepted to Interspeech 2025. - Researched neural vocoders for high-quality speech generation from self-supervised speech features; the work was accepted to ICASSP 2026.	Japan Dec 2024 – Mar 2026 1 year 4 months
NTT Communication Science Laboratories , Research Intern Researched post-refinement of ASR hypotheses for joint multi-channel distant speech recognition.	Japan Feb 2025 – Mar 2025 1 month
LY Corporation R&D , Research Intern Improved phonemic and prosodic annotation models for text-to-speech and related subtasks.	Japan Aug 2024 – Oct 2024 3 months
The University of Tokyo , Research Intern Researched environmental sound synthesis from visual onomatopoeia; the work was accepted to ICASSP 2023.	Japan June 2022 – July 2022 2 months

Education

Ph.D. The University of Osaka , Engineering Science - Graduate School of Engineering Science, Doctoral Course. Dialogue Intelligence Systems Laboratory (Yoshino Laboratory), supervised by Prof. Koichiro Yoshino.	Japan Oct 2026 – present
M.S. Nara Institute of Science and Technology , Information Science - Graduate School of Science and Technology, Master's Course. - Intelligent Robot Dialogue Laboratory (RIKEN), supervised by Prof. Koichiro Yoshino. - Received the NAIST Best Student Award.	Japan Apr 2024 – Mar 2026
B.E. National Institute of Technology, Tokuyama College , Computer Science and Electronic Engineering - Advanced Course. - Miyazaki Laboratory, supervised by Prof. Ryoichi Miyazaki.	Japan Apr 2022 – Mar 2024
National Institute of Technology, Tokuyama College , Computer Science and Electronic Engineering - Regular Course. - Miyazaki Laboratory, supervised by Prof. Ryoichi Miyazaki.	Japan Apr 2017 – Mar 2022

Selected Publications

- Wave-Trainer-Fit: Neural Vocoder with Trainable Prior and Fixed-Point Iteration towards High-Quality Speech Generation from SSL Features** May 2026
Hien Ohnaka, Yuma Shirahata, Masaya Kawamura
[10.1109/ICASSP55912.2026.11461239](https://doi.org/10.1109/ICASSP55912.2026.11461239) (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP))
- Grapheme-Coherent Phonemic and Prosodic Annotation of Speech by Implicit and Explicit Grapheme Conditioning** Aug 2025
Hien Ohnaka, Yuma Shirahata, Byeongseon Park, Ryuichi Yamamoto
[10.21437/Interspeech.2025-661](https://doi.org/10.21437/Interspeech.2025-661) (Interspeech)
- Modeling Turn-Taking Speed and Speaker Characteristics** Aug 2025
Kazuyo Onishi, Hien Ohnaka, Koichiro Yoshino
aclanthology.org/2025.sigdial-1.2 (Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL))
- Unsupervised Speech Enhancement with Spectral Kurtosis and Double Deep Priors** Feb 2025
Hien Ohnaka, Ryoichi Miyazaki
doi.org/10.1250/ast.e24.66 (Acoustical Science and Technology)
- Visual Onoma-to-Wave: Environmental Sound Synthesis from Visual Onomatopoeias and Sound-Source Images** June 2023
Hien Ohnaka, Shinnosuke Takamichi, Keisuke Imoto, Yuki Okamoto, Kazuki Fujii, Hiroshi Saruwatari
doi.org/10.1109/ICASSP49357.2023.10096517 (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP))

Awards

- Best Student Award** Mar 2026
Awarded by Nara Institute of Science and Technology. [Announcement](#)
- Committee Special Award** Mar 2025
Awarded by the Association for Natural Language Processing for work on jointly predicting response timing and short responses for real-time spoken dialogue systems. [Award details](#)
- IEEE SPS Tokyo Joint Chapter Student Award** Dec 2024
Awarded for Visual Onoma-to-Wave. [Award details](#)

Domestic Presentations (Non-peer-reviewed)

- 音声の意味的類似度に基づいた対照学習によるリアルタイム応答選択** Mar 2026
大中 緋慧, 大西 一誉, 吉野 幸一郎
(言語処理学会第 32 回年次大会)
- Voice Activity Projection とガンマ分布ハザード関数を用いて対話の間を考慮したターンシフトモデル** Mar 2026
大西 一誉, 大中 緋慧, 吉野 幸一郎
(言語処理学会第 32 回年次大会)
- リアルタイム音声対話システムのための応答タイミングと短文応答の同時予測** Mar 2025
大中 緋慧, 河野 誠也, 大西 一誉, 吉野 幸一郎
www.anlp.jp/proceedings/annual_meeting/2025/pdf_dir/P10-4.pdf (言語処理学会第 31 回年次大会)

話者特性に基づくターンテイキング速度の分析 大西 一誉, 大中 緋慧, 吉野 幸一郎 www.anlp.jp/proceedings/annual_meeting/2025/pdf_dir/D8-1.pdf (言語処理学会第 31 回年次大会)	Mar 2025
情報案内・傾聴対話における対話行為とターン交替時間の関係分析 大中 緋慧, 河野 誠也, 大西 一誉, 吉野 幸一郎 doi.org/10.11517/jsaislud.102.0_96 (人工知能学会研究会資料 言語・音声理解と対話処理研究会)	Nov 2024
F0 に基づいて伸縮された画像文字からの音声合成 大中 緋慧, 宮崎 亮一, 高道 慎之介 speakerdeck.com/nehio615/f0niji-duiteshen-suo-saretahua-xiang-wen-zi-karanoyin-sheng-he-cheng (日本音響学会講演論文集)	Mar 2024
事前学習不要な DNN 音声強調の反復処理に基づく性能改善 中田 優翔, 広政 遼汰, 大中 緋慧, 宮崎 亮一 (日本音響学会講演論文集)	Mar 2024
クリーン不要な DNN 音声雑音除去に用いる目的信号の分析 広政 遼汰, 大中 緋慧, 宮崎 亮一 (日本音響学会講演論文集)	Mar 2024
過学習の抑制に基づく事前学習不要な DNN 音声強調 中田 優翔, 広政 遼汰, 大中 緋慧, 宮崎 亮一 (第 26 回関西支部若手研究者交流研究発表会)	Dec 2023
Double-DIP に基づく教師なし雑音除去の分析と性能改善 大中 緋慧, 宮崎 亮一 (日本音響学会講演論文集)	Mar 2023
Visual onoma-to-wave：音源画像と画像オノマトペを利用した環境音合成の提案 大中 緋慧, 高道 慎之介, 井本 桂右, 岡本 悠希, 藤井 一貴, 猿渡 洋 (電子情報通信学会 音声研究会)	Feb 2023
クリーン音声を必要としない時間領域での DNN 音声強調 広政 遼汰, 大中 緋慧, 宮崎 亮一 (第 25 回関西支部若手研究者交流研究発表会)	Nov 2022
Double-DIP に基づく単一雑音混入音声のみを用いた雑音除去の検討 大中 緋慧, 藤村 拓弥, 宮崎 亮一 (電子情報通信学会 応用音響研究会)	May 2022
深層ニューラルネットワークにおける学習過程の信号分布に着目した雑音抑圧の検討 大中 緋慧, 藤村 拓弥, 宮崎 亮一 (第 24 回関西支部若手研究者交流研究発表会)	Dec 2021